

Ion Trap versus Low-Energy Beam-Type Collision-Induced Dissociation of Protonated Ubiquitin Ions

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The beam-type and ion trap collision-induced dissociation (CID) behaviors of protonated bovine ubiquitin ions were studied for charge states ranging from +6 to +12 on a modified triple quadrupole/linear ion trap tandem mass spectrometer. Both beam-type CID and ion trap CID were conducted in a high-pressure linear ion trap, followed by proton-transfer ion/ion reactions to reduce the charge states of product ions mostly to +1. The product ions observed under each activation condition were predominantly b- and y-type ions. Fragmentation patterns showed a much stronger dependence on parent ion charge state with ion trap CID than with beam-type CID using nitrogen as the collision gas, with preferential cleavages C-terminal to aspartic acid at relatively low charge states, nonspecific fragmentation at moderate charge states, and favored cleavages N-terminal to proline residues at high charge states. In the beam-type CID case, extensive cleavage along the protein backbone was noted, which yielded richer sequence information (77% of backbone amide bond cleavages) than did ion trap CID (52% of backbone amide bond cleavages). Collision gas identity and collision energy were also evaluated in terms of their effects on the beam-type CID spectrum. The use of helium as collision gas, as opposed to nitrogen, resulted in CID behavior that was sensitive to changes in collision energy. At low collision energies, the beam-type CID data resembled the ion trap CID data with preferential cleavages predominant, while at high collision energies, nonspecific fragmentation was observed with increased contributions from sequential fragmentation.

The ability to study large biomolecules via mass spectrometry has been greatly enhanced by the development of electrospray ionization (ESI)¹ and matrix-assisted laser desorption/ionization (MALDI).² Formation and mass analysis of large gas-phase biomolecule ions with molecular masses up to hundreds of megadaltons is now possible.³ Furthermore, biomolecule structure

information can be obtained in a rapid and sensitive fashion with the use of tandem mass spectrometry. In the case of proteins, there are two classes of tandem mass spectrometry approaches to obtaining primary structure information for the purpose of protein identification and characterization. The most widely used class takes a so-called “bottom-up”⁴ approach, which involves protein digestion either before or after separation. After the separation and digestion steps, the resultant peptides are ionized and subjected to tandem mass spectrometry^{5,6} to provide peptide primary structure information that can be related back to the original proteins. While the bottom-up methodology is widely recognized for its effectiveness in protein identification, limitations, such as the loss of the intact protein mass information, congestion of peaks in a narrow mass range, and discrimination of certain peptides from ionization of complex peptide mixtures, warrant the development of complementary or alternative approaches. Therefore, an alternative class of approaches, which does not entail protein digestion, is being developed and explored. This class of approaches is often referred to as “top-down”, because intact protein ions are subjected to tandem mass spectrometry and the masses of the protein and its fragment ions are used for protein identification.⁷ The advantages inherent to top-down approaches include the preservation of intact protein mass information and the direct linkage between molecular mass and the structural information derived from tandem mass spectrometry. The first experiment employing a top-down approach was performed on a triple quadrupole instrument by Loo et al. in the early 1990s.^{8,9} However, ambiguity in the assignment of fragment ion charge due to the inability of the mass analyzer to resolve isotope peaks limited the utility of the approach for protein characterization. To date, most top-down applications have been carried out using Fourier transform ion cyclotron resonance (FTICR) mass spectrometers^{10,11} or on tandem mass spectrometers that employ time-

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of-flight mass analyzers.¹² These approaches rely on the resolution of product ion isotope spacing for the determination of fragment charge. An alternative approach to dealing with product ion charge-state determination has been to manipulate product ion charge using ion/ion reactions. This approach has allowed ion trap instruments to be utilized for whole protein structural characterization.

A key element in all top-down strategies is the means by which structural information is derived. All approaches attempted to date rely on dissociation of protein ions in the gas phase. The dissociation behavior of protein ions is less well understood than that of peptide ions, but several factors are known to play important roles, depending upon the conditions under which dissociation is induced. For example, the charge state of the precursor ion^{13–18} can be a major variable in determining the favored dissociation channels when the ion trap activation method is used. For a given charge state, the primary sequence (e.g., the presence of particular amino acids, sometimes with other key amino acids adjacent to them)^{19–22} as well as secondary structure¹⁷ can play major roles. Important characteristics of the means for inducing dissociation include the time window within which dissociation reactions occur, the internal energy distribution of the activated ions, and the mechanism by which energy is transferred into the protein. A variety of activation methods have been employed to effect fragmentation of multiply protonated protein ions in the gas phase, such as collision-induced dissociation (CID),^{9,12,14,23} including nozzle–skimmer dissociation,^{24,25} and the sustained off-resonance irradiation^{26,27} approach in FTICR instruments, ultraviolet photodissociation,²⁸ blackbody infrared radiative dissociation,¹⁸ and infrared multiphoton dissociation.²⁹ A distinct approach to protein ion dissociation involves either the capture by the protein ion of a low-energy electron or the transfer of an electron from an anion to the protein ion. These techniques are referred to as electron capture dissociation^{30,31} and electron-transfer dissociation,³² respectively. Of all the approaches used

to dissociate protein ions, collisional activation, in one form or another, is available with essentially all commercial tandem mass spectrometers, whereas the others are not. For this reason, it is important to fully characterize this approach so that it can be maximally exploited.

The CID spectra of ions are often sensitive to collisional activation conditions such as collision energy, target gas identity, target gas pressure, and the observation window of the tandem mass spectrometer. In terms of collision energy, CID in beam-type tandem mass spectrometers can be divided into two regimes, the high-energy regime (keV/charge, laboratory collision energies) and the low-energy regime (collision energies typically less than 100–200 eV/charge). Among common tandem mass spectrometers, high collision energies are accessible with TOF/TOF and sector instruments, although the latter are rarely used for whole protein analysis. Tandem time-of-flight instruments have been used to study whole protein dissociation behavior, although all such studies to date have been restricted to the use of MALDI as the ionization method and, as a consequence, have largely focused on singly charged ions.^{33–35} While the data reported to date show that high-energy CID of singly charge proteins can yield useful structural information, richer structural information has generally been obtained from multiply charged ions activated at lower collision energies.

Within the category of low-energy CID, conditions can be subdivided further on the basis of the time frame of activation. A common low-energy approach, referred to as low-energy beam-type CID, directs a beam of ions into a collision region where collisional excitation takes place on the time frame of the passage of the ions through the collision cell. Furthermore, the observation window in most instruments that employ this approach (e.g., triple quadrupole and quadrupole-TOF instruments) requires that dissociation takes place at rates greater than $\sim 10^4 \text{ s}^{-1}$, which is consistent with the residence times of ions in the collision cell. In the case of ion-trapping instruments, ion activation/deactivation occurs continuously and at comparable rates during an ion activation period of tens to hundreds of milliseconds. This approach tends to sample dissociation rates in the range of 1–100 s^{-1} . Several proteins have been systematically studied with ion trap CID as a function of parent ion charge state.^{14,15,19,36–39} In these studies, common trends have been identified for the dissociation behavior of various protein ions under ion trap CID conditions, such as neutral losses and cleavage C-terminal to aspartic acid

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residues at low charge and nonspecific cleavages at intermediate charge. The dissociation behavior at high charge states shows the least commonality between different proteins, but it is not uncommon to observe dominant N-terminal proline cleavages at high charge states (although such cleavages are also sometimes prominent at lower charge as well). It has been found that additional specificity in protein identification can be obtained by ranking matches to favored cleavages, such as C-terminal Asp and N-terminal Pro cleavages, more highly than others. Relatively little systematic data on whole protein CID under low-energy beam-type CID conditions have been reported. Given the significantly different conditions used to effect beam-type CID, it is of interest to study the fragmentation behavior of protein ions under such conditions and to compare the observed behavior with ion trap CID. Of particular interest is the extent to which structural information obtained via the different approaches overlaps as well as the utility of the data in making confident protein identifications.

In this paper, the dissociation behavior of ubiquitin ions has been studied as a function of charge state via two dissociation methods, ion trap CID and beam-type CID, in a high-pressure linear ion trap. For the beam-type activation method, the identity of collision gas and the collision energy were further evaluated in terms of their effects on the beam-type CID spectrum. The beam-type CID experiments were conducted under conditions in which the ions were allowed to accumulate in the linear ion trap, as opposed to being transmitted directly through a mass-resolving quadrupole. Hence, products from dissociation reactions that took place over a wide range of rates were integrated. This approach is more efficient than the transmission mode of operation and has the advantages for this work of providing a time window similar to that for ion trap CID for collection of CID products, as well as subjecting the product ions to identical mass analysis conditions for both activation techniques.

EXPERIMENTAL SECTION

Bovine ubiquitin was purchased from Sigma-Aldrich (St. Louis, MO). Perfluoro-1-octanol (PFO) was obtained from Sigma-Aldrich (Milwaukee, WI). Samples were used without further purification. Solutions of ubiquitin were dissolved to 5 μ M in 50/50/1 (v/v/v) methanol/water/acetic acid to yield charge states from +8 to +12 or in pure water solution to yield +6 and +7 charge states. PFO, with a final concentration of 100 μ M, was prepared in methanol solution containing 1% ammonium hydroxide.

Ionization was accomplished by a home-built pulsed dual electrospray source,⁴⁰ which comprised a nanoelectrospray emitter for the formation of various charge states of ubiquitin cations and an electrospray emitter for the formation of deprotonated PFO anions. The nanospray solutions were loaded into borosilicate glass capillaries (1.55-mm o.d., 0.86-mm i.d.) pulled by a P-87 Flaming/Brown micropipet puller (Sutter Instruments, Novato, CA). A stainless steel wire was inserted into the back of the capillary to apply high voltage. The PFO solutions were introduced using a Harvard 11 syringe pump (Harvard Apparatus Inc., Holliston, MA) at a flow rate of 1 μ L/min into a fused-silica capillary (100- μ m i.d., 200- μ m o.d.). The position of each sprayer can be adjusted separately and was tuned to achieve the best

signal response. High voltages applied to each emitter were optimized separately and were independently generated by a high-voltage power supply from the instrument and a fast high-voltage pulser (GRX-1.5K-E, Directed Energy Inc., Fort Collins, CO). The two power supplies were triggered so that the high voltages were alternately switched on (off) in each scan. The "off" voltages for each emitter were 0 V, and the "on" voltages were between 1.2 and 2 kV for the nanoelectrospray emitter and -2.5 kV for electrospray.

All experiments were performed on a prototype version of a Q TRAP mass spectrometer⁴¹ (Applied Biosystems/MDS Sciex, Concord, ON, Canada) that had been modified for ion/ion reactions.⁴² The Q TRAP was operated at a drive rf of 650 kHz with a nominal upper mass-to-charge limit of 4200 Th, when the standard mass-selective axial ejection (MSAE)⁴³ frequency at 247 kHz was used. The auxiliary rf signals applied to IQ2 and IQ3 lenses were optimized and set at 290 kHz, 100 V_{0-p} for each ion/ion reaction experiment. Instrument control and data acquisition were carried out using MS Expo 1.78 software, provided by MDS Sciex.

For a beam-type CID experiment, the parent ions selected by Q1 in the mass-resolving mode were accelerated into the pressurized Q2 collision cell (nitrogen, $\sim$ 7 mTorr). For each charge state of ubiquitin, spectra were collected for at least three different collision energies. Helium has been evaluated as collision gas for selected charge states. To perform ion trap CID in Q2, Q1 selected precursor ions were injected into Q2 at relatively low kinetic energy ($Q0-Q2 = 4$ V) to minimize fragments caused by beam-type CID. After ions were translationally cooled, a dipolar rf signal with frequency in resonance with the parent ions was applied across one set of opposing rods of the Q3 quadrupole array for 300 ms to induce ion trap CID in Q2 (the Q2 and Q3 quadrupole arrays are capacitively coupled). The resonance excitation amplitudes were adjusted for each charge state of ubiquitin to optimize the product ion signal-to-noise ratio. Product ions generated either from beam-type CID or ion trap CID were cooled in Q2 for 50 ms after the excitation process. In the ensuing anion injection step, the negative high voltage for the formation of deprotonated PFO ions was triggered on for 50 ms, while the dc potentials applied to the ion path prior to Q2 were adjusted to inject PFO anions into Q2 at low kinetic energies and the dc potentials on IQ2 and IQ3, as well as on the Q2 rods themselves, were ramped to the same potential level. To prevent ions from leaking out of the Q2 cell, an auxiliary rf signal was applied to the IQ3 lens during this anion injection step. In the subsequent mutual ion storage step, the auxiliary rf signals were applied to both IQ2 and IQ3 to store ions axially. After 150 ms of mutual storage, application of the auxiliary rf signals was terminated and the reagent anions were ejected from Q2 by applying 200 V dc potentials to the IQ2 and IQ3 lenses. The product ions were then transferred from Q2 to Q3 and stored for 50 ms before they were subjected to MSAE using a supplementary rf signal at frequency 247 kHz, or at 84 kHz to give the desired mass range extension.⁴⁴ The spectra

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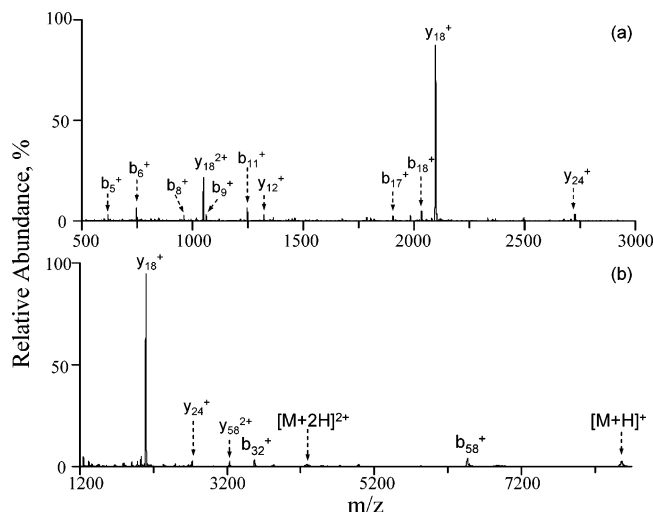


Figure 1. Post-ion/ion ion trap CID spectra of the $[M + 6H]^{6+}$ ion of ubiquitin via activation at 45.43 kHz, 1.5 V_{0-p} for 300 ms. (a) Low-mass range spectrum collected at ejection frequency 247 kHz and (b) high-mass range spectrum collected at 84 kHz.

shown here were typically the average of 100–200 individual scans. The post-ion/ion MS/MS spectra were externally calibrated with the +1, +2, and +3 charge states of ubiquitin ions via ion/ion reactions. A fragment ion mass tolerance window of ± 2 Da was used in the manual assignment of product ions.

RESULTS AND DISCUSSION

Ion Trap CID. Protonated ubiquitin ions ranging in charge states from +6 to +12 were subjected to ion trap collisional activation in the Q2 LIT, followed by ion/ion proton-transfer reactions with deprotonated PFO anions to reduce most of the product ions to the +1 charge state. The proton-transfer ion/ion reactions serve two purposes in the study of the product ion spectra. First, by converting all product ions largely to the +1 charge state, many possible charge-state ambiguities associated with the initially formed product ions are avoided. Second, the relative contributions of various dissociation channels can be evaluated without considering charge-state-dependent ion detection efficiency. However, it is noted here that the detection system shows higher responses for low-mass singly charged ions than for high-mass singly charged ions. Post-ion/ion MS/MS spectra for each collision condition were acquired over two mass-to-charge ranges, m/z 500–3000 and 1200–9000, because a single MSAE frequency could not be used to collect data over the entire mass-to-charge range of interest. An example is shown in Figure 1, which illustrates fragmentation observed for the $[M + 6H]^{6+}$ ion of ubiquitin under ion trap activation conditions (dipolar excitation at 45.43 kHz, 1.5 V_{0-p} , 300 ms, nitrogen as collision gas at 7 mTorr) in the Q2 LIT after a 150-ms ion/ion proton-transfer reaction period. Since most of the ions are reduced in charge to +1 via ion/ion reactions, product ions are readily assigned by comparing observed m/z values with expected singly charged product masses of ubiquitin. Fragmentation of 11 backbone amide bonds is identified with y_{18}^+ as the most dominant fragment ion, which is formed from cleavage C-terminal to an aspartic acid residue (Asp⁵⁸). A clearer picture of the amide cleavages that contribute to the MS/MS data is provided by summing the abundances of the complementary b- and y-type ions and plotting the totals,

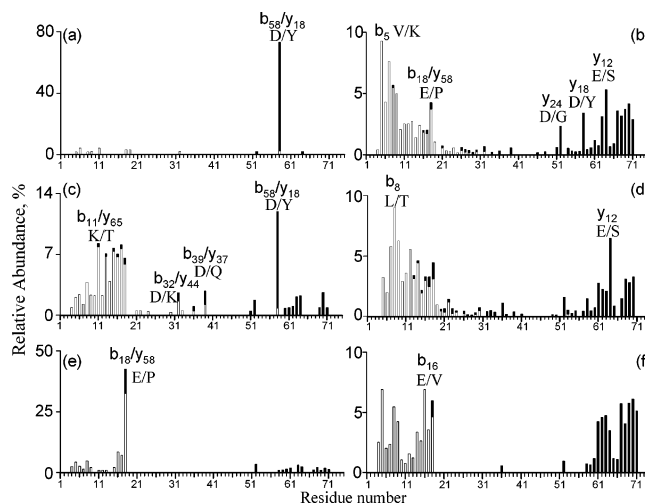


Figure 2. Relative abundances of b-type (white column) and y-type (black column) product ions resulting from cleavage of the ubiquitin backbone with ion trap CID for (a) $[M + 6H]^{6+}$, (c) $[M + 8H]^{8+}$, and (e) $[M + 11H]^{11+}$ ions; and with beam-type CID for (b) $[M + 6H]^{6+}$, (d) $[M + 8H]^{8+}$, and (f) $[M + 11H]^{11+}$ ions.

expressed as a fraction of total product ion abundance, versus residue number. To relate abundances noted in the low mass range spectrum to those in the high mass range spectrum, the ion abundances were normalized on the basis of an ion that appears in both spectra. In the case of the $[M + 6H]^{6+}$ ion, the y_{18}^+ ion served this purpose whereas the b_{18}^+ ion did so for the other charge states. The product ions with at least 3% of the abundance of the most prominent product ion were included in the summed b- and y-ion abundance plot and were converted into relative abundances by normalizing to the total abundances of all assigned product ions. The contribution from the b-ion abundance to the bar associated with cleavage at a given amide bond is shown in white and the y-ion abundance is shown in black. The post-ion/ion MS/MS spectra shown in Figure 1 for the $[M + 6H]^{6+}$ ubiquitin ion were thus transformed to Figure 2a and similarly for all other charge states.

The ion trap fragmentation behavior of ubiquitin ions from $[M + 6H]^{6+}$ to $[M + 12H]^{12+}$ in the Q2 LIT was found to be very similar to that observed previously using a 3-D ion trap.¹⁶ That is, preferential fragmentation C-terminal to aspartic acid residues at low charge state (e.g., +6), relatively nonspecific fragmentation at intermediate charge states (+7 to +9), and a high propensity for cleavage N-terminal to proline residues at the highest charge states (+10 to +12). The 3-D ion trap data were collected using helium as the target gas at a pressure of 1 mTorr, whereas these data were collected using nitrogen as the target gas at a pressure of 7 mTorr. The resonance excitation amplitudes, however, were also different, while similar time frames were employed. It is likely, therefore, that the effective temperatures of the ions were similar. The general trend in dissociation behavior is reflected by the product ion abundance plots of $[M + 6H]^{6+}$, $[M + 8H]^{8+}$, and $[M + 11H]^{11+}$ as shown in Figure 2a, c, and e, respectively. For the ion trap CID of the $[M + 6H]^{6+}$ parent ion, more than 70% of the total product ion abundance was accounted for by cleavages C-terminal to aspartic acid residues. However, the contribution from C-terminal aspartic acid cleavage decreased to $\sim 20\%$ in the product ion abundances of $[M + 8H]^{8+}$, with the appearance of

more evenly weighted nonspecific fragmentation channels. For even higher charge states, +11 for example, the favored fragmentation channel shifted to cleavage N-terminal of Pro¹⁹, which accounted for 42% of the total product ion abundance. Similar charge-state-dependent fragmentation behavior has been observed for a variety of proteins and peptides following ion trap collisional activation and is consistent with notions of proton mobility playing a major role.²² When the excess protons are strongly bound and have limited mobility, those mechanisms that require mobile protons contribute little whereas those processes that do not require mobile protons tend to be prominent, such as cleavage C-terminal to aspartic acid residues. For the intermediate charge states, at least some of the ionizing protons are sufficiently weakly bound to be somewhat mobile in the activated ions so that they can catalyze amide backbone cleavages, thereby giving rise to nonspecific fragments, in addition to those favored fragmentation channels observed for the low charge states. The preferential cleavage N-terminal to proline in peptides has been accounted for by the conformational influence of the proline residue on protonation and subsequent fragmentation.^{45,46} However, the high basicity of the tertiary amide bond of the proline residue has also been discussed within the context of charge-directed fragmentation.^{47,48} The generally lower diversity of fragmentation observed at high charge states, relative to intermediate charge states, is also consistent with limited proton mobility. In this case, proton mobility can be limited by the relatively high electrostatic field of the ions.¹⁷

Beam-Type CID. The beam-type CID conducted in a triple quadrupole instrument differs mainly in the activation time frame and resulting internal energies of precursor ions, when compared to ion trap collisional activation, which is one of the slow heating methods.⁴⁹ Higher collision energies (e.g., up to 200 eV/charge), relative to ion trap collisional activation, are typically accessible in beam-type experiments conducted in a triple quadrupole or quadrupole-TOF instrument. Ion translational energies in ion traps are limited by trapping well depths, which, under typical collisional activation conditions in ion traps, are less than 10 eV/charge.⁵⁰ Activation occurs by continuous acceleration in the presence of the bath gas, typically resulting in over 1000 collisions. A higher steady-state internal energy distribution is sustained, which, when the amplitude is sufficiently high, results in the high-energy tail of the distribution exceeding critical energies for ion dissociation at rates of 1–100 s⁻¹.⁵¹ To drive dissociation at rates commensurate with the time scale of a conventional beam-type instrument (i.e., generally in excess of 10⁴ s⁻¹), ions must be excited to energies higher than those typically encountered in an ion trap collisional activation experiment. In the data collection approached here, where product ions are accumulated in the LIT while they are collisionally activated by injection at relatively high translational energies, the integration time over which products

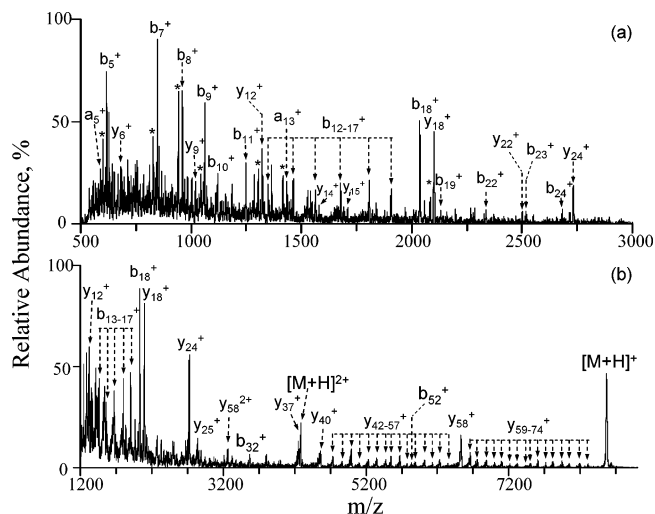


Figure 3. Post-ion/ion beam-type CID spectra of the $[M + 6H]^{6+}$ ion of ubiquitin: collision energy, 444 eV; collision gas, nitrogen at 7 mTorr. (a) Low-mass range spectrum collected at ejection frequency 247 kHz and (b) high-mass range spectrum collected at 84 kHz. The loss of ammonia or water from the corresponding a, b, or y ions are labeled with an asterisk.

are collected is similar to that for a typical ion trap experiment. However, the beam-type experiment also integrates products formed at significantly higher dissociation rates, which is the origin of the differences in the data collected by the two approaches (i.e., ion trap resonance excitation versus high-energy injection into the LIT).

The charge states of ubiquitin studied via ion trap CID were subjected to beam-type CID in the Q2 LIT, followed by proton-transfer ion/ion reactions to simplify the product ion spectra. The collision energy of each charge state of ubiquitin was adjustable by changing the potential difference of the Q2 rod offset relative to the Q0 rod offset. Nitrogen was used as the collision gas and was maintained at a pressure of ~7 mTorr. This pressure resulted in good parent ion to product ion conversion as well as high ion/ion reaction rates. At this pressure, the mean free path between collisions is roughly 0.02–0.05 cm for ions with collision cross sections of 1000–2000 Å², values that apply to ubiquitin ions.⁵² Given the length of Q2 quadrupole array (12.7 cm),⁴¹ multiple collisions are thus involved in the beam-type experiments, similar to the ion trap collisional activation experiments. However, the collision energies and time frames differ under the two activation methods, and it is these differences that likely underlie the differences in dissociation behavior. As with the ion trap CID experiments, two post-ion/ion MS/MS spectra were acquired for each collision condition to cover both low-mass and high-mass regions. The post-ion/ion beam-type CID spectra of the $[M + 6H]^{6+}$ ions are shown as an example in Figure 3 (CID gas, N₂; pressure, 7 mTorr; collision energy, 444 eV; laboratory frame of reference). The product ions with an ion intensity of ≥3% of the most prominent product ion were selected and assigned as a-, b-, or y-type ions by comparison with the expected product ion masses of ubiquitin. Extensive fragmentation was observed as evidenced by the observation of a-type ions and the loss of

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ammonia or water from the corresponding a-, b-, or y-type ions, which are labeled with an asterisk in Figure 3. Compared with the ion trap CID of the $[M + 6H]^{6+}$ ubiquitin ions (Figure 1), the beam-type CID data clearly show a much richer array of fragmentation. For example, consecutive b ions, from b_7 to b_{19} , are obvious in Figure 3a, while consecutive y ions, from y_{42} to y_{74} , are shown in Figure 3b. For this particular charge state, cleavage of 52 out of 75 backbone amide bonds was observed, presenting a large contrast to the ion trap CID of the same charge state (11 out of 75).

For various charge states of the ubiquitin ions, the relative contributions of fragment channels in the post-ion/ion beam-type CID were plotted with regard to their residue numbers, as described earlier for the ion trap CID data. Direct comparisons of each charged state under different CID conditions, e.g., ion trap CID versus beam-type CID, but under identical mass analysis conditions, are therefore readily made. The fragmentation patterns of the $[M + 6H]^{6+}$, $[M + 8H]^{8+}$, and $[M + 11H]^{11+}$ ions are shown in Figure 2b, d, and f, respectively, and can be compared to corresponding ion trap CID data in Figure 2a, c, and e. It is clear from the comparisons of Figure 2 that there are similarities as well as dramatic differences in the ion trap versus beam-type CID data. Both forms of ion activation lead largely to the formation of b- and y-type fragments. Both sets of data show significant contributions from first-generation cleavages, although there is much more evidence for sequential cleavages in the beam-type CID data (see below, as well). Furthermore, cleavage for this particular protein is largely present at the termini. That is, most ions fragment near to the N- and C-termini of the protein and few tend to fragment in the middle. It is clear in Figure 2 that the favored fragmentation channels in the ion trap CID data, such as cleavages C-terminal to aspartic acid residues in $[M + 6H]^{6+}$ and the cleavage N-terminal to Pro¹⁹ in $[M + 11H]^{11+}$, are also evident in the beam-type CID of the corresponding charge states, but the relative contributions from those channels are greatly reduced. This is because there are increased contributions from nonspecific fragmentation. The difference is dramatic for the comparisons of the $[M + 6H]^{6+}$ and $[M + 11H]^{11+}$ ions and less so for the comparison of the $[M + 8H]^{8+}$ ions, the ion trap CID data for which show the most extensive nonspecific fragmentation. It is apparent that the beam-type CID results show much less sensitivity to parent ion charge over the range of charge states studied. For the $[M + 6H]^{6+}$ and $[M + 11H]^{11+}$ ions in particular, beam-type activation leads to many more fragmentation channels and highlights the sensitivity of ion trap collisional activation conditions (i.e., relatively low energies and long times) to small differences in threshold dissociation behavior.⁴⁹

Another obvious difference of the dissociation patterns between beam-type and ion trap CID is that more C-terminal end fragments are found in the beam-type CID than in the ion trap CID of various charge states of ubiquitin. In the previous study of the dissociation behavior of ubiquitin ions in a 3-D ion trap, notice was drawn to the modest contributions from fragmentation in the C-terminal half of ubiquitin.¹⁶ It was reasoned that the presence of the four arginine residues of ubiquitin in the C-terminal side of the protein might lead to higher relative proton sequestration in that region of the protein than in the N-terminal side. Wysocki et al. have reported that higher energies are required for proton mobiliza-

tion from arginine side chains, a prerequisite to peptide backbone cleavage from charge-directed mechanisms.²² The data are therefore consistent with greater internal energy deposition in the beam-type CID experiment, which might be expected to lead to greater proton mobilization both in the various charge states and in the C-terminal end of the protein. The higher internal energies likely acquired by the precursor ions under beam-type activation, as well as the multiple-collision nature of the experiment, which give rise to activation of first-generation products, also lead to greater contributions from sequential fragmentation. It has been shown that further activation of y-type ions derived from first-generation ubiquitin product ions yields a series of smaller y ions.¹⁶ The presence of more abundant small y-type ions in the beam-type CID data may arise, at least in part, from the sequential dissociation of larger y ions. Sequential dissociation in single-frequency ion trap resonance excitation tends to be minimized as the first-generation products are not subject to ion acceleration and tend to be cooled by collisions with the bath gas.

Although the fragmentation patterns in the beam-type CID experiments show less sensitivity to parent ion charge than the ion trap CID data, differences do exist from low charge states to high charge states. For example, in low charge states (+6 to +8), more cleavages were observed in the middle of the protein backbone chain than noted in the higher charge states (+9 to +12). It is tempting to ascribe this observation to inhibition of proton mobility due to the electrostatic field within the ion that results from multiple charging. The highest charge states of proteins derived from ESI are commonly observed to show a more limited array of cleavages than intermediate charge states under slow heating ion activation conditions. Low charge states also show cleavages at limited numbers of amide bonds, typically C-terminal to aspartic acid residues. This observation has been attributed to low proton mobility due to protonation at highly basic sites, such as arginine, and to intramolecular charge solvation. These factors apparently do not play as a dominant role in the case of beam-type CID, but limited proton mobility at higher charge states due to electrostatic effects may show a greater influence. Clearly, studies with other proteins are warranted to determine the generality of these observations.

Effects of Collision Energy and Collision Gas in Beam-Type CID of Ubiquitin. Collision energy, target gas mass, and target gas pressure are the three main experimental variables associated with beam-type CID. In these studies, target pressure was not a readily adjustable variable over a wide enough range to allow for reliable systematic studies due to the need for relatively high pressures for efficient ion/ion reactions. Collision energy, on the other hand, was readily variable and the use of helium as a target gas is straightforward with the present instrumentation. The roles of collision energy and target gas mass are apparent in the consideration of the kinematics associated with a single collision. The motion of the center of mass in a binary collision is conserved, such that only a fraction of the collision energy in the laboratory frame of reference is available for partitioning into internal energy of the collision partners. This energy is referred to as the collision energy in the center of mass frame of reference, KE_{com} , and is related to the laboratory collision energy (KE_{lab}) by the relation given by eq 1:

$$KE_{\text{com}} = \left(\frac{N}{m_p + N} \right) KE_{\text{lab}} \quad (1)$$

where N is the mass of the neutral target and m_p is the mass of the ion.⁵³

In most beam-type CID experiments, the target gas can be assumed to be at rest such that the kinetic energy of the precursor ions largely determines the laboratory collision energy. In the present instrumentation, the potential difference between the Q0 and Q2 quadrupoles determines the parent ion kinetic energy as it enters Q2. For each charge state of ubiquitin ranging from +6 to +12, the kinetic energy of the precursor ions was varied to study how the collision energy affected the beam-type CID spectra. Interestingly, the CID spectra collected using nitrogen as the collision gas showed very little sensitivity to collision energy once the collision energies were high enough to induce dissociation. Apparently, at some point beyond the dissociation threshold, the relative rates of cleavage of the various competing dissociation channels become insensitive to internal energy. The range of collision energies over which collision energy dependence can be observed is apparently quite narrow with nitrogen as the target gas at 7 mTorr. Therefore, it was reasoned that collision energy effects might be more readily studied by reducing the energy-transfer step size in the multiple collision beam-type CID experiment via reduction of the mass factor in eq 1, i.e., $N/(m_p + N)$.

Helium was evaluated as the collision gas for beam-type activation because its use reduces the mass factor in eq 1 by a factor of 7 relative to use of nitrogen. The pressure of Q2 was set at ~10 mTorr, which provided the highest transmission and sensitivity, for the studies of beam-type CID of $[M + 10H]^{10+}$ ubiquitin ions described below. The collision energy of the $[M + 10H]^{10+}$ ubiquitin ions was stepped at 200 eV (20 V/charge) increments from 600 to 1400 eV. The relative abundances of fragments varied little in the high-mass region (m/z 2000–9000) of the post-ion/ion spectra, while significant differences were observed in the low-mass region (m/z 500–2000), as demonstrated in Figure 4a and b. When the collision energy was relatively low, e.g., 600 eV (Figure 4a), preferential cleavage N-terminal to Pro¹⁹ gave rise to b_{18}^+ as the most abundant product in the low-mass region. Fragments from cleavages adjacent to Pro¹⁹, such as b_{16}^+ and b_{17}^+ , were also abundant with all other fragments contributing much less. When the energy was increased to 1200 eV, as shown in Figure 4b, the relative abundance of b_{18}^+ ions was only 20% that of the y_{13}^+ ions, which arise from cleavage between Lys⁶³–Glu⁶⁴. Fragments from cleavages at residues adjacent to this site were also prominent, such as the y_{12}^+ , y_{14}^+ , and y_{15}^+ ions. Other fragmentation channels noted in the higher energy CID data include the cleavages between Leu⁶⁹–Val⁷⁰ and Val⁵–Lys⁶. Compared to the lower energy CID data, more sequence ions were observed in the high-energy spectrum than in the low-energy data. It is also clearly apparent that contributions from sequential fragmentation that does not lead to b- and y-type ions were also much greater in the high collision energy data, which can complicate spectral interpretation. The post-ion/ion data over the same m/z range from ion trap CID (dipolar excitation at, 76.9 kHz; amplitude, 2.5 V_{0-p} , 300 ms) and

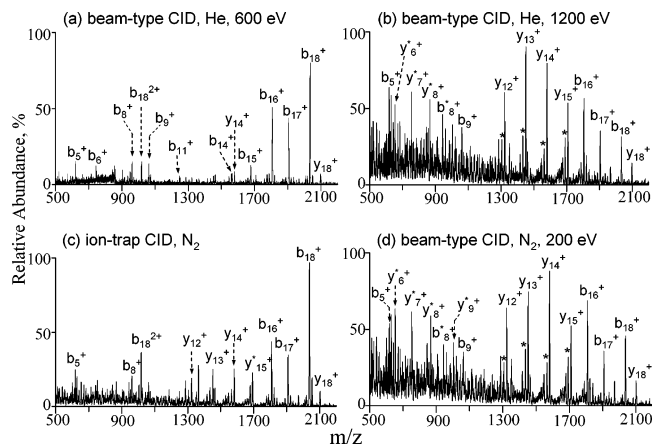


Figure 4. Post-ion/ion data of ubiquitin $[M + 10H]^{10+}$ ions under (a) beam-type CID: collision gas, helium; collision energy, 600 eV; (b) beam-type CID, collision gas, helium; collision energy, 1200 eV; (c) ion trap CID, collision gas, nitrogen; excitation at 76.9 kHz; activation amplitude, 2.5 V_{0-p} ; activation time, 300 ms; and (d) beam-type CID, collision gas, nitrogen; collision energy, 200 eV.

beam-type CID (200 eV) using nitrogen as collision gas are illustrated in Figure 4c and d, respectively. Striking similarity can be found between the spectra of ion trap CID (bath gas, nitrogen) and beam-type CID (bath gas, helium, 600 eV). In addition, the spectrum of beam-type CID with nitrogen as bath gas (200 eV) is almost a duplicate of the spectrum collected for the 1200-eV beam-type CID with helium as bath gas.

These observations demonstrate that the appearance of beam-type product ion spectra is more variable with helium as collision gas than with nitrogen, presumably due to the lower slope (by a factor of 1/7) associated with the relationship between KE_{com} and KE_{lab} . Therefore, the differences found between ion trap CID and beam-type CID when using nitrogen as collision gas are mostly due to the different internal energies deposited in the precursor ions. Once the energy differences are small, the CID spectra for beam-type versus ion trap CID are very similar. The method of collecting product ions is probably important in this regard. By accumulating product ions prior to scanning, the time window for collection of beam-type CID products is similar to that for an ion trap CID experiment. Hence, it is possible to collect CID products from the relatively low-energy ions activated by beam-type CID with helium because the product ion collection period is sufficiently long to accumulate ions from the relatively low rate dissociations that contribute to the data.

The effect of collision energy in the beam-type CID of +10 ubiquitin ions is summarized in Figure 5, which shows the CID data collected with collision energies of 600 (Figure 5a), 1000 (Figure 5b), and 1400 eV (Figure 5c), respectively, in which product ion abundances are plotted as a function of residue number. Only the fragment ions in the low-mass region (m/z 500–2000) are plotted and normalized to the most prominent product ion in Figure 5, since the relative abundance of each fragment varied little in the high-mass region (m/z 2000–9000). The b ions are represented by white bars whereas the relative abundances of products from sequential fragmentation, such as (b-NH₃/H₂O) ions, a ions, and (a-NH₃/H₂O) ions, are summed and plotted as black bars. The y ions and their ammonia or water loss ions are also shown in each plot by gray and striped gray bars, respectively.

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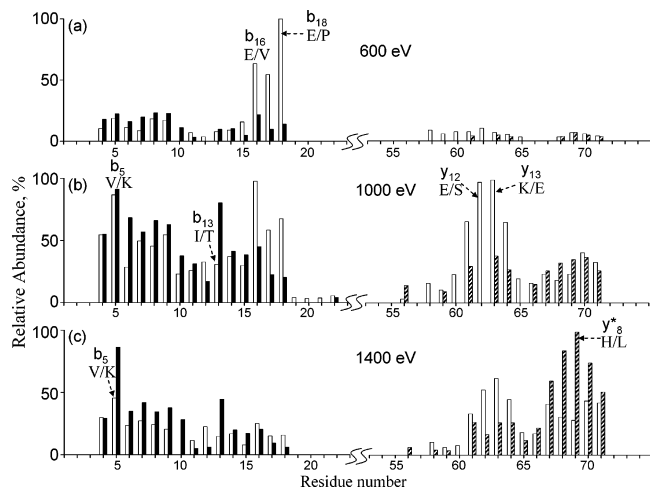


Figure 5. Relative abundances of product ions resulting from beam-type CID of ubiquitin $[M + 11H]^{11+}$ ions with helium as collision gas under different precursor ion collision energies: (a) 600, (b) 1000, and (c) 1400 eV. The relative abundances of b-type ions are represented by white columns. The sums of (b-NH₃/H₂O) ions, a ions, and (a-NH₃/H₂O) ions are indicated by black columns. The relative abundances of y-type ions and their ammonia or water loss ions are shown in gray and striped gray columns, respectively.

The effect of collision energy on the abundances of b- and y-type ions and their sequential dissociation products can be clearly seen in Figure 5. When the energy is low (600 eV), facile fragmentation channels such as cleavages between Glu¹⁶–Val¹⁷ and Glu¹⁸–Pro¹⁹ are prominent, while the C-terminal end fragments (small y ions) are barely seen, in analogy with ion trap CID results. When the collision energy is increased to 1000 eV, new fragmentation channels open up, such as the cleavages between Lys⁶³–Glu⁶⁴, Glu⁶⁴–Ser⁶⁵, and Val⁵–Lys⁶. However, these fragmentation channels are joined by other cleavages as major channels when the kinetic energy is further increased to 1400 eV, with the His⁶⁸–Leu⁶⁹ cleavage, for example, representing one of the predominant fragmentation channels. In addition to changes in the relative abundances of b- and y-type ions, increasing collision energy also leads to an increase in the relative abundances of ions from sequential cleavages. That is, the secondary ions from b- or y-type ions are formed with increasing abundances when the energy is increased. However, at the highest collision energy (1400 eV), the relative contributions from many of the b-type ions and their summed secondary ions appear to decrease. The latter observation may be due to further sequential fragmentation that gives rise to products that are not recognized as such (e.g., internal fragments) or are not collected, either due to scattering or due to formation of sequential products below the low-mass cutoff of Q2.

Ion Trap versus Beam-Type CID for Protein Identification and Characterization. The observations related here hold significance for top-down protein identification and characterization because it is clear that beam-type CID can provide structural information that exceeds that obtained via ion trap CID, at least for ubiquitin. The percentage of amide bond cleavages noted for the various charge states of ubiquitin ions under beam-type CID (nitrogen as collision gas at 7 mTorr) and ion trap CID is summarized in Table 1.

Note that only b- or y-type ions of abundance higher than 3% of the most abundant product ions were included. Higher degree

Table 1. Percentage of Amide Bond Cleavages Observed for Ubiquitin Ions under Ion Trap CID and Beam-Type CID Conditions

charge state	amide bond cleavages, %		
	ion trap	beam type	combined
+6	14.7	69.3	69.3
+7	40.0	68.0	70.6
+8	45.3	68.0	72.0
+9	45.3	57.3	58.7
+10	41.3	57.3	61.3
+11	33.3	52.0	52.0
+12	36.0	54.6	56.0

of protein backbone cleavages might be achieved with a lower threshold, but this presumes that the signal-to-noise ratios of such low-abundance products are sufficiently high for confident assignment. The 3% abundance value was selected as a practical threshold. For ion trap CID data, the highest degree of backbone cleavages (~40%) was noted for the intermediate charge states (+7 to +10) with lower values for both low and high charge states. Generally much more protein backbone cleavages were found for all the charge states in the beam-type activation experiments, with the highest (~70%) found for relatively low charge states (+6 to +8). Less cleavages (~50%) were seen for high charge states (+9 to +12), possibly due to reduced proton mobility, as discussed above. The backbone cleavages from all the charge states of the ubiquitin ions resulting from ion trap CID and beam-type CID are summarized in Charts 1 and 2, respectively. A total of 39 (52%) backbone cleavages are found for the ion trap CID, while 58 (77%) cleavages are identified for beam-type CID. Even higher percentages of backbone cleavages for ubiquitin ions under beam-type CID conditions have been reported recently, such as 98% via nozzle–skimmer ion activation²⁵ and ~90% under beam-type activation on a quadrupole TOF instrument.⁵⁴ Several factors may contribute to the observed differences in the degree of backbone cleavages in these approaches. However, it is likely that most of the differences can be accounted for on the basis of the criteria for including an ion with those used to make structural assignments (e.g., the inclusion of internal fragments²⁵ or a different threshold criterion). In any case, more backbone cleavages of ubiquitin ions are noted for all of the variations of beam-type collisional activation than is observed with ion trap CID. Comparing Charts 1 and 2, it is clear that essentially all cleavages that appear in ion trap CID also appear in the beam-type CID. For this reason, combining the fragment information from ion trap CID to the beam-type CID data only marginally increases the number of backbone cleavages, as shown in the right-hand column of Table 1.

From the standpoint of protein identification and characterization, higher degree of backbone cleavages is more desirable because the chance for random matches to a product ion spectrum decreases dramatically as the number of predicted fragments that are observed increases.⁵⁵ Furthermore, a wider variety of cleavages along the protein backbone allows, in principle, for a higher

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Chart 1. Combined Backbone Cleavages Resulting from the Ion Trap CID of Ubiquitin Ions

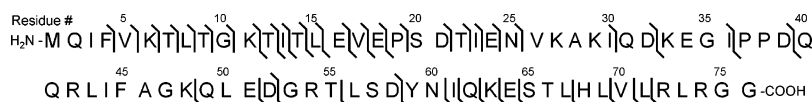
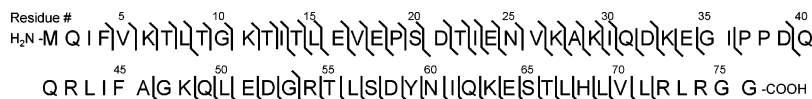


Chart 2. Combined Backbone Cleavages Resulting from the Beam-Type CID of Ubiquitin Ions



likelihood for localization of posttranslational modifications. It is also advantageous to be able to derive as much sequence information as possible from a single charge state. Such a scenario can avoid the need to acquire MS/MS spectra from multiple charge states.³⁸ Therefore, a single ion parking⁵⁶ step into any charge state, for example, followed by beam-type CID could provide essentially all of the primary structural information obtainable via collisional activation. In this case, there would be minimal need (or little use) for repeating the process with ion parking into a different charge state. These considerations favor the use of beam-type CID with nitrogen as the target gas over ion trap CID (or beam-type CID using helium as the target and low collision energies) as the means for deriving primary sequence information.

Given the foregoing discussion, it is worth considering if there are any circumstances under which it might be more desirable to use ion trap CID (or beam-type CID at low collision energies with helium as the target) rather than beam-type CID. Only a general commentary can be provided at this point. The advantages described above for beam-type CID derive from the fact that it is a less specific activation method, in terms of the cleavages that it induces. The question can then be formulated on the basis of the specificity of the activation method. In terms of minimizing detection limits, it is generally desirable to minimize the number of fragment channels, provided there is sufficient information present for protein identification/characterization. Furthermore, if it is desirable to employ several stages of ion activation, as in an MSⁿ procedure, the number of stages of MS is limited by the number of different product channels into which the precursor signal for each stage is partitioned.⁵⁷ For these reasons, an activation method that leads to relatively few fragmentation channels, but ones that provide useful structural information, can complement a nonspecific activation method. Ion trap CID of relatively low charge states that show selective cleavages C-terminal to aspartic acid residues has the characteristics just mentioned. This tendency has been noted for many proteins and is the closest analogy observed to date between gas-phase protein cleavage and specific enzymatic or chemical cleavages in the condensed phase. Recognition of this specificity has allowed for the use of scoring algorithms that weight such cleavages more strongly and provide improved specificity in distinguishing between candidate proteins in a database search. Hence, the studies reported here suggest that beam-type CID of whole protein ions has highly useful characteristics for protein identification, relative

to ion trap CID, due to its relatively nonspecific nature. However, ion trap CID of relatively low charge states may provide useful complementary information due to its tendency for inducing cleavages C-terminal to aspartic acid residues.

CONCLUSIONS

The dissociation behavior of whole protein ions in the gas phase is highly dependent upon ion activation conditions. In this study, efforts were directed to the comparison of collisional activation methods applied to various charge states of ubiquitin. Specifically, the two forms of collisional activation, ion trap activation and beam-type activation, were applied to ubiquitin ions with charge states ranging from +6 to +12. The comparisons were made using similar time windows for product ion accumulation and detected under identical mass analysis conditions. For the ion trap activation method, which employed nitrogen as the target gas, the observed fragmentation behavior was very similar to that noted in a previous study using ion trap collisional activation in a 3-D ion trap with helium as the bath gas. Ubiquitin ions showed fragmentation behavior similar to that commonly observed with other proteins. Specifically, at relatively low charge states (e.g., +6), preferential cleavages C-terminal to aspartic acid residues were observed. At relatively high charge states (e.g., +10 to +12), cleavages N-terminal to proline were dominant. With the presence of such facile fragmentation channels, less sequence information was retrieved from these charge states, when compared to the dissociation behavior of the intermediate charge states, where more nonspecific fragmentation was observed. For beam-type activation using nitrogen as the target gas, extensive fragmentation of ubiquitin ions was noted with higher degree of backbone cleavages for all the charge states than for any charge state subjected to ion trap collisional activation. The spectra were much less sensitive to parent ion charge state than those acquired using ion trap collisional activation. Furthermore, relatively little sensitivity to collision energy was noted. However, the beam-type CID spectra were more sensitive to collision energy when helium was used as the collision gas. When the collision energy was low, spectra that resembled the ion trap CID data were observed. With an increase in collision energy, more channels were represented in the spectra, to the point where the data resembled that acquired using nitrogen as the collision gas. Hence, a wide range of dissociation phenomenology is accessible with beam-type collisional activation, depending upon collision energy and target gas identity. The comparison with ubiquitin ions indicates that beam-type collisional activation holds particular utility for protein identification and characterization, due to its relatively nonspecific nature and its relative insensitivity to parent ion charge state.

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These characteristics make it complementary to the much more specific dissociation observed with ion trap collisional activation, particularly at low and high charge states. The highly promising results noted for beam-type collisional activation indicate that further comparisons of this type with other proteins are warranted to determine the generality of the observations made in this work.

ACKNOWLEDGMENT

This work was sponsored by MDS Sciex, an Industrial Associate of the Department of Chemistry. The authors acknowl-

edge technical support from Dr. Robert E. Santini and Mr. Mark Carlsen of the Jonathan Amy Facility for Chemical Instrumentation. Y.X. acknowledges the support from Merck Research Laboratories Fellowship during part of the project.

Received for review September 10, 2005. Accepted November 18, 2005.

AC051622B